Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/18/2006
PI Name:	Hermanson, James C Ph.D.		
Project Title:	STABILITY AND FLOW STRUCTURE OF EVAPORATING FILMS IN REDUCED GRAVITY		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jherm@aa.washington.edu	**Fax:**	FY 206-543-0217
PI Organization Type:	UNIVERSITY	**Phone:**	206-616-2310
Organization Name:	University of Washington		
PI Address 1:	Department of Aeronautics and Astronautics		
PI Address 2:	1100 NE Campus Pkwy, Box 352400		
PI Web Page:	http://www.aa.washington.edu/		
City:	Seattle	**State:**	WA
Zip Code:	98195-6605	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Griffin, DeVon	**Contact Phone:**	216-433-8109
Contact Email:	DeVon.Griffin@grc.nasa.gov , DeVon.W.Griffin@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 09/30/2007 per E. Golliher/GRC (7/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Allen, Jeffrey (Michigan Technological University) Bailey, Michael (University of Washington)		
Grant/Contract No.:	NNC04GA76G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This experimental study of film evaporation in normal- and reduced gravity involves three experimental configurations. The first fluid configuration is an evaporating film on a heated substrate. This configuration is expected to yield Marangoni and/or Rayleigh-Bénard convection in reduced gravity, with those behaviors modified by the action of evaporation. In the second configuration, heat is provided directly to the liquid surface, for example by infrared heating. In this case evaporation is present but in the near absence of the forces that drive Marangoni and/or Rayleigh-Bénard convection. In the third configuration, a non-evaporating, heated film is studied. In this case thermocapillary effects will potentially be present owing to the heating of the film, but effects of mass removal and vapor recoil due to vaporization are removed. Reduced-gravity experiments will be conducted on board NASA's parabolic-trajectory aircraft; experiments are also conducted in the laboratory in normal gravity. Experimental diagnostics include detailed temperature and heat transfer measurements as well as ultrasonic measurement of film thickness. In addition, optical imaging is employed to determine the features of the resulting instabilities and associated changes in the structure of the film. The experiments are supplemented by stability analysis of fluid layers undergoing evaporation. The objective of this research is to investigate the fundamental physics of film evaporation in reduced gravity. The evaporation rate, film stability, and interfacial motion are studied by isolating the effects due to surface tension gradients, buoyancy, latent heat, and mass flux. This objective is being met using normal gravity experiments, reduced gravity experiments on NASA's Reduced Gravity Aircraft, and through stability analysis. The evaporation of liquid on a heated surface is a phenomenon of importance in many engineering problems involving heat transfer and phase change. There has been relatively little research on the detailed mechanisms operative at a film interface undergoing evaporation in reduced-gravity conditions. Of particular interest and significance are the stability characteristics of the film, and how they differ from those of films of comparable scale in the absence of evaporation. Mechanisms that can impact the stability of evaporating films include thermocapillary mechanisms that drive instabilities and fluid motion in liquid layers with non-uniform temperature, as well as the vapor recoil associated with the mass loss due to evaporation. Both of these effects are expected to play a de-stabilizing role for film layers in reduced-gravity, where surface-tension-driven convection can become important. The high heat flux associated with phase change is an important consideration in spacecraft thermal control applications. Devices such as capillary pumped loops and heat pipes depend critically on evaporation and the successful utilization of these devices in space applications require a detailed understanding of evaporation phenomena under reduced gravity conditions. Other important technologies related to space exploration requiring a more in-depth understanding of evaporation include humidity control of cabin environment, water recovery from waste sources, oxygen production, water management in power systems such as fuel cells, and in-situ resource utilization for fuel production. Understanding evaporation phenomena is critical to the development and optimization of these space technologies that enable human exploration of space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In addition to space applications, evaporation is important to many terrestrial technologies from drying applications in the pulp and paper industry to coating deposition on fiber optics. Many of these technologies, while reliable, are not optimized for lack of a thorough understanding of evaporation phenomena. International competition is driving a need beyond consistent, reliable operation of manufacturing and processing technologies towards more optimized operations. Research on evaporation mechanisms may assist in this optimization. The use of ultrasound for the dynamic monitoring of film thickness and growth rate developed as part of this effort has potentially widespread use for the determination of film behavior in a variety of devices that employ two-phase flow, such as power stations and closed-loop refrigeration cycles, evaporators, and condensers.
	Evaporation and condensation experiments have been conducted in normal gravity in a test cell containing a downwards-facing, cooled brass block with an exposed diameter of 10 cm. Vapor is provided by evaporating n-pentane in a 500 ml degassed Pyrex container. The temperature at the exposed surface of the brass block is regulated by a circulating temperature-controlled water/glycol mixture flowing through a serpentine channel at the opposite end of the block. A vacuum pump is used to evacuate air from the test cell and to ensure positive flow of vapor through the test cell during evaporation and condensation. The metal surface is polished to allow for double-pass shadowgraph imaging. An embedded thin film heat flux sensor measures the heat flux and temperature without disturbing the fluid film. The sensor is ringed by a film of Kapton of identical thickness and thermal conductivity as the heat flux gauge to ensure uniform thermal resistance across the cross section of the test chamber. Thermal data are acquired using a PC-based system, which also grabs the video images of the film to allow for synchronization of the shadowgraph imaging with the temperature and heat flux measurements. Film growth under both evaporating and condensing conditions is examined by imposing a cyclic variation in the system pressure. This allows for examination of the full range of evaporating/condensing film behavior, ranging from thin films, unstable films, pendant drops, evaporating films, to film dry-out. The pressure cycling is accomplished using an electronic pressure controller. A one-dimensional, finite-difference, inverse heat conduction algorithm has been developed to calculate the unsteady heat flux at the surface using data acquired by the embedded heat transfer sensor. The numerical method divides the heated/cooled block into two regions, a direct region between the sensor location and the heat source/sink plane (the circulating liquid) with known temperature, and an indirect region between the sensor location and the surface at which the liquid film resides. The solution for the temperature and heat flux in the direct region solution is straightforward. An inverse method is used in the indirect region. The heat flux input to the algorithm uses a non-sinusoidal curve fit to heat flux data over a typical condensation/evaporation cycle. The accuracy of the inverse method was verified by comparison with known temperatures and heat fluxes for the case of a dry surface. Validation experiments have been conducted for unstable (i.e. surface facing downwards) condensing films under steady-state conditions for system pressure ranging from 60 kPa to 75 kPa and subcooling of up to 5 C. The measured heat transfer under steady-state conditions results in Nusselt numbers in agreement with previously published correlations for steady, unstable films. For unsteady tests of evaporating and condensing films, the heat flux and the surface temperature vary substantially over the course of each cycle. The heat flux evidently increases most dramatically due to the initial formation of a thin film of condensate in the condensation portion of each cycle. Although the measured heat flux increases with increasing

Task Progress:	vapor pressure during each cycle, the spatially averaged heat flux does not appear to be significantly impacted by the appearance and amplification of Rayleigh-Taylor instability waves at the surface of the growing film during condensation. Similar behavior is seen for all test cases examined to date. For cycles with the lowest difference between the vapor and surface temperatures, increases in heat flux due to natural convection prior to the development of the condensate film are observed. For larger temperature differences this phenomenon is less apparent, being quickly overwhelmed by the onset of condensation. The greater the peak difference between the vapor and surface temperatures, the more cycles are required to achieve a quasi-steady state, cyclic condition. A substantial delay (on the order of 10 seconds) between changes in the measured heat flux at the sensor and that calculated for the surface is predicted. This long delay time appears to be in large part related to the additional thermal resistance in the solid block at the location of the embedded probe. For the case of cyclic pressure variation a degree of hysteresis has been observed, with the heat flux and surface temperatures being different during the evaporation and condensation portions of the cycle for a given system pressure. The heat flux for these unsteady conditions thus appears to be influenced by more than the degree of superheating or subcooling of the evaporating or condensing film, respectively. Other factors that impact the heat flux in unsteady evaporation and condensation include variations in the film thickness (including the formation of pendant drops during condensation) and the thermal mass of the solid wall material. Preparations are underway for conducting evaporation and condensation experiments in reduced gravity on board NASA's parabolic-trajectory aircraft. A special test cell to rapidly initiate evaporation and condensation for aircraft experiments has been developed and tested. Rapid heat transfer is achieved by several thermoelectric coolers in conjunction with a pressure/volume control system. The test cell is a 12 cm diameter, 25 cm long glass cylinder with copper plates at the bottom and at the side. The remaining end consists of a window to allow for optical access to the evaporating/condensing film. The surface at which the evaporating/condensing films reside is 2.5 cm thick to maintain a nearly constant wall temperature. The other plate (0.64 cm thick) is rapidly heated or cooled to provide a vapor source/sink as needed to minimize the required volume change during film evaporation and condensation. Successful operation of the thermoelectric heating/cooling system has been demonstrated. A pressure/volume control system to quickly achieve and maintain the desired system pressure required to initiate evaporation and condensation under reduced-gravity conditions has been designed. An ultrasound measurement system has been developed to measure the temporally- and spatially-varying evaporating film thickness in both normal and reduced-gravity experiments. The configuration consists of a Panametrics-NDT model 25MX Plus pulser/receiver generator combined with a GE Panametrics model MX8 multiplexer. The three transducers are Panametrics-NDT models M110-RM (6 mm diameter, 5.0MHz), M112-RM (6 mm diameter 10.0MHz), and M116-RM (3 mm diameter 20.0Mhz). Effort this year has been directed at improving the spatial resolution of the thickness measurement by each individual transducer, and to balance bandwidth versus sensitivity to obtain a measurable signal of very short duration for optimal spatial resolution. Successful testing of stable (upwards-facing) condensing, evaporating, and boiling films has been accomplished. Acoustic signals have been generated and propagated through a 2.54 cm-thick copper plate and reflecting at both interfaces of a stagnant, non-evaporating water film of known thickness resting on the top surface of the plate. Film thicknesses as small as 50 to 100 mm have been resolved to date, with a resolution of roughly 10 to 20 mm and measurement times on the order of 10 ms. The software-controlled measurement system is currently being modified to allow the use of an array of multiple transducers. Activity and progress over the next three months of the current reporting period is expected to include the following: First, additional experiments will be conducted to examine the impact of different thermal boundary conditions on the hysteresis in heat flux and film temperature during unsteady evaporation and condensation. Additional validation tests of the ultrasound system for different transducers and operating frequencies will be followed by detailed measurement of the dynamic film thickness under the unsteady conditions that occur during cyclic evaporation/condensation experiments. The mechanical and optical design of the aircraft rig for reduced-gravity testing is expected to be completed, to be followed by fabrication and testing. Numerical analysis of unsteady film behavior will be conducted based on finite difference techniques where the film thickness variation is analyzed using the nonlinear evolution equation for films with phase change. The numerical results will be obtained for distinct cases of fluid films, such as isothermal films and quasi-equilibrium and non-evaporating films. The findings from numerical analysis will be useful in determining the control parameters for the experiments to be conducted on NASA's Reduced Gravity Aircraft.
Bibliography Type:	Description: (Last Updated: 09/03/2025)
Articles in Peer-reviewed Journals	Zhen ZQ, Hermanson JC, Shear MA, Pedersen PC. "Ultrasonic monitoring of interfacial motion of condensing and non-condensing liquid films." Flow Measurement and Instrumentation 2005 Dec;16(6):355-64. http://dx.doi.org/10.1016/j.flowmeasinst.2005.06.002 , Dec-2005